Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			**Indicative content:** **Apparatus set-up:** Set up the clamp. Suspend the spring from the clamp. Place a ruler in a [vertical] position alongside the spring. **Method:** Record the original length of the spring. Attach the 100 g mass hanger. Record the length of the spring with the 100 g mass hanger. Add a further 100 g to the spring and record the new length. Repeat until all masses have been added. Take repeat readings. **Analysis:** Calculate the mean length for each mass added. Calculate the extension for each mass added. Plot a graph of force / mass against extension. **5–6 marks** Description of the apparatus set-up, method and analysis. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3–4 marks** Description of 2 out of 3 of the apparatus set-up, the method or the analysis **or** a limited description of all 3. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.*	6			6		6

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Description of 1 out of 3 of the apparatus set-up, the method or the analysis **or** a limited description of 1 or 2 areas. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
	(b)	(i)		15 [cm] **ecf** can't be applied for an answer of 0 here in the next 2 parts		1		1		1
		(ii)		Substitution: $F = 0.8 \times 15$ (**ecf**) (1) = 12 [N] (1)	1	1		2	2	2
		(iii)		Mass = $\frac{12(\textbf{ecf})}{10}$ (1) = 1.2 [kg] (1)	1	1		2	2	2
				Question 4 total	**8**	**3**	**0**	**11**	**4**	**11**